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Engineering®

V Semester

Artificial Intelligence and Machine Learning (IS353IA) Open-Ended Project Work

Department of Computer Science & Engineering,
RV College of Engineering, Bengaluru.

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Traffic Monitoring System Using YOLO and Deep SORT

Chandan Kumar K R	1RV23CS402
Gopinath Ramaje	1RV23CS404
KIRAN H R	1RV23CS405

Faculty In charge (Theory) :

Dr.Soumya. A
Associate Professor

Faculty In charge (Lab) :

Dr.Sindhu D V
Associate Professor



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Agenda

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- Identification of Problem Domain and Detailed Analysis
- Literature Review
- Research gap
- Objectives
- Methodology
- Design
- Implementation



Problem Domain

- The rapid growth of urbanization has led to a significant increase in vehicular traffic, making efficient traffic monitoring essential for ensuring road safety and effective traffic management.
- Traffic congestion, road safety, and law enforcement are significant challenges in urban transportation systems.
- Traditional traffic monitoring methods rely on fixed sensors, manual observations, or GPS-based tracking, which often lack real-time adaptability and accuracy. The emergence of **computer vision and deep learning** provides an opportunity to develop a more efficient and scalable traffic monitoring system that can **detect, track, and analyze vehicle movement in real time**.



Potential for AI in the Problem Domain:

- AI-powered traffic monitoring can automate and optimize traffic management using object detection, tracking, and predictive analytics. This technology can be integrated into smart city infrastructure

Problem statement:

- Current traffic surveillance methods (like fixed sensors, manual observations, or GPS-based tracking) suffer from **high costs, inefficiency, and lack of real-time adaptability**.



Purpose of the Project

The primary objective of this project is to **enhance real-time traffic monitoring** using AI, improving both efficiency and accuracy. The key goals include:

- **Automating traffic surveillance** to minimize human intervention.
- **Accurately estimating vehicle speed** using **Bird's Eye View Transformation** and **frame rate-based calculations**.
- **Detecting and classifying vehicles** to provide insightful traffic data

Uniqueness of the Project

Unlike traditional traffic monitoring systems, this project integrates **advanced AI-based tracking and speed estimation techniques**.

- Bird's Eye View Transformation for Accurate Speed Estimation
- Real-Time Multi-Object Tracking with Deep SORT
- Yolo for classifying vehicle



Sl. No	Title of the work	Author/s and Publication details	Objectives of the work carried out	Results Obtained	Gaps Identified
1	Real-Time Traffic Management Using Deep Learning and Object Detection using YOLOv8	Denis Pato Martínez et al., Advances in Intelligent Systems and Computing, Aug 2022	Proposed a deep learning system to detect traffic density using YOLO.	Achieved varying detection accuracy based on training datasets	Effectiveness depends on dataset quality; suggests need for comprehensive datasets.
2	Intelligent Infrastructure for Traffic Monitoring Based on Deep Learning	Hindawi, 2024	Developed a real-time traffic monitoring solution using edge computing and existing cameras.	Enhanced monitoring without additional sensors or halting traffic.	Scalability and integration with current systems not addressed.



Sl. No	Title of the work	Author/s and Publication details	Objectives of the work carried out	Results Obtained	Gaps Identified
3	AI-Powered Traffic Monitoring and Analysis with YOLO	International Journal of Advanced Research in Computer and Communication Engineering, 2024	Implemented YOLOv8 for real-time object detection and traffic analysis.	Provided high accuracy in traffic analysis.	Performance under varying conditions not discussed.
4	Real-Time Traffic Monitoring using Computer Vision and Deep Learning	International Research Journal of Engineering and Technology, 2024	Introduced a method for real-time traffic detection using deep learning and computer vision.	Developed an effective real-time monitoring system.	Lacks detailed performance evaluation and limitations in diverse conditions.



Sl. No	Title of the work	Author/s and Publication details	Objectives of the work carried out	Results Obtained	Gaps Identified
5	Visual Detection of Traffic Incidents through Automatic Monitoring of Urban Environments	<i>World Electric Vehicle Journal</i> , 2024	Combined YOLOv8 with Deep SORT to monitor urban traffic and detect incidents.	Effectively detected incidents and monitored traffic patterns..	Effectiveness in non-urban settings and scalability not explored.



objectives:

1. **Vehicle Detection and Tracking:** Use YOLOv8 for object detection and Deep SORT for tracking vehicles (cars and trucks). Implement real-time detection and tracking to monitor each vehicle's movement.
2. **Speed Calculation:** Implement a method to calculate vehicle speed based on their movement across a transformed bird's-eye view of the road, using the frames per second (FPS) and the distance traveled by each vehicle.
3. **Ticket Generation:** Automatically generate tickets for speeding vehicles, based on predefined thresholds (e.g., 120 km/h for cars and 100 km/h for trucks). Capture images of the vehicles at the moment of speeding.
4. **Save Processed Data:** Store images of speeding vehicles and their associated information in a directory. Output the processed video, which includes FPS tracking and vehicle annotations.



1. Initialize Video Capture, YOLOv8 Model, and Deep SORT Tracker

- Set up video input and output, load the YOLOv8 model for vehicle detection, and initialize the Deep SORT tracker to maintain vehicle identity across frames.

2. Detect and Track Vehicles with YOLOv8 and Deep SORT

- Use YOLOv8 to detect vehicles in each frame, and apply Deep SORT to assign and track unique IDs for each vehicle, maintaining consistent tracking across frames.

3. Apply Bird's Eye View Transformation

- Transform the road view to a bird's-eye view to accurately calculate the vehicle's speed and movement on a flat plane.

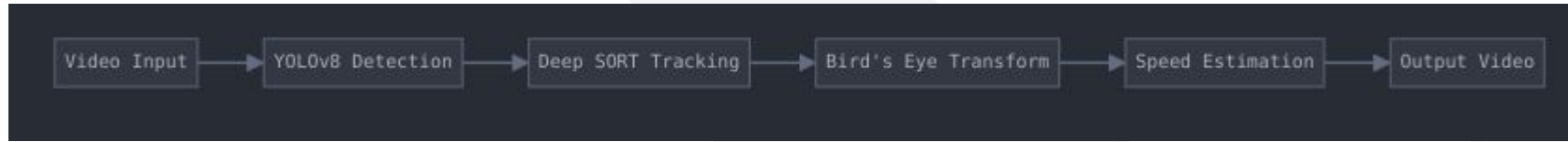


4. Calculate Vehicle Speed Using Kalman Filter

- Use the Kalman Filter to predict and smooth the vehicle's position over time, enhancing the speed calculation accuracy.

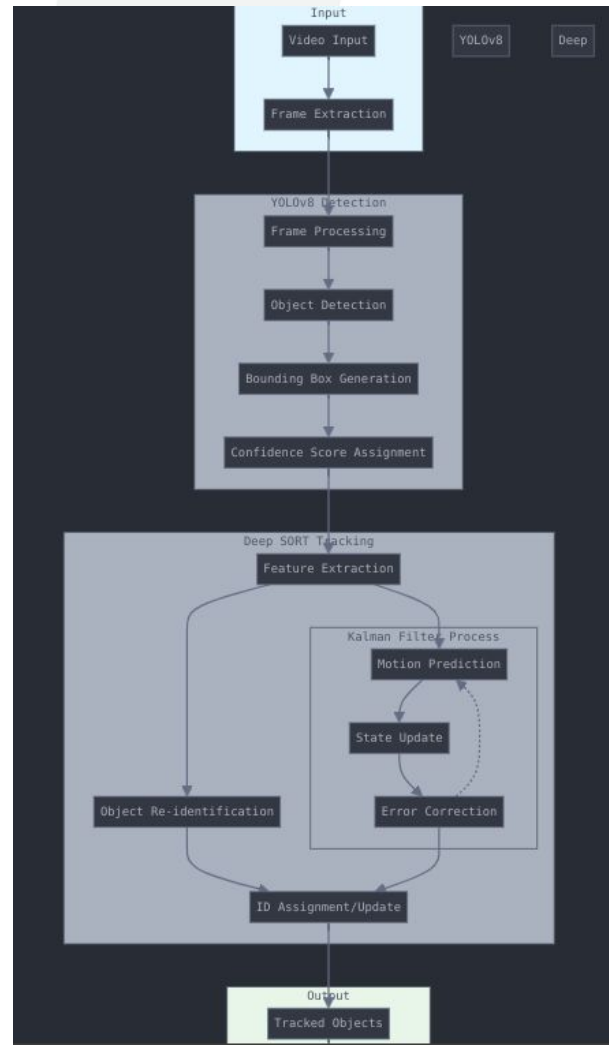
5. Generate Ticket for Speeding Vehicles

- Identify speeding vehicles based on the speed calculations, capture images, and generate a PDF ticket with vehicle details and speed, including the logo.





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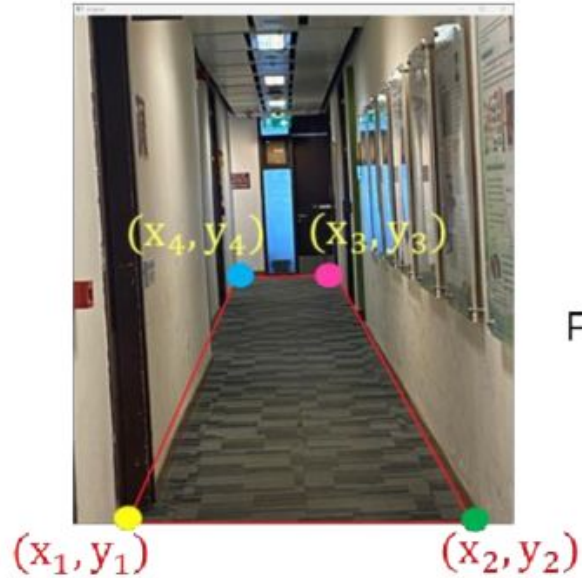
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Bird's Eye View Transformation

- When tracking vehicles on the road using a normal camera, perspective distortion makes it difficult to measure distances accurately. Vehicles closer to the camera appear larger, and those further away appear smaller.
- To solve this, we use a Bird's Eye View (BEV) transformation, also called perspective transformation, which converts the road scene from a camera perspective to a top-down view.

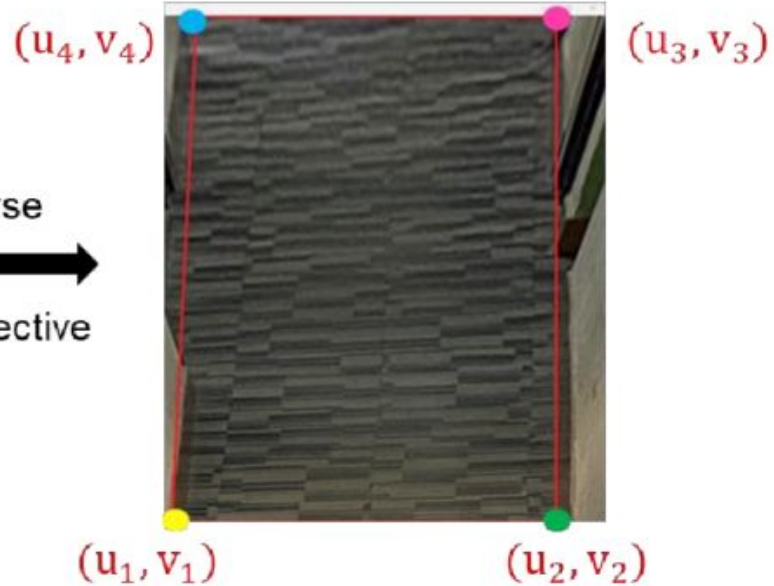
Perspective view



Inverse

 Perspective

Bird's eye view





Speed Calculation:

1. Measuring Pixel-Based Distance

- Once a vehicle's bottom-center position is mapped to the bird's eye view, we store its y-coordinates over time in a dictionary.
- The difference between consecutive y-coordinates gives the pixel distance traveled.
- Larger movements in consecutive frames indicate higher speed.

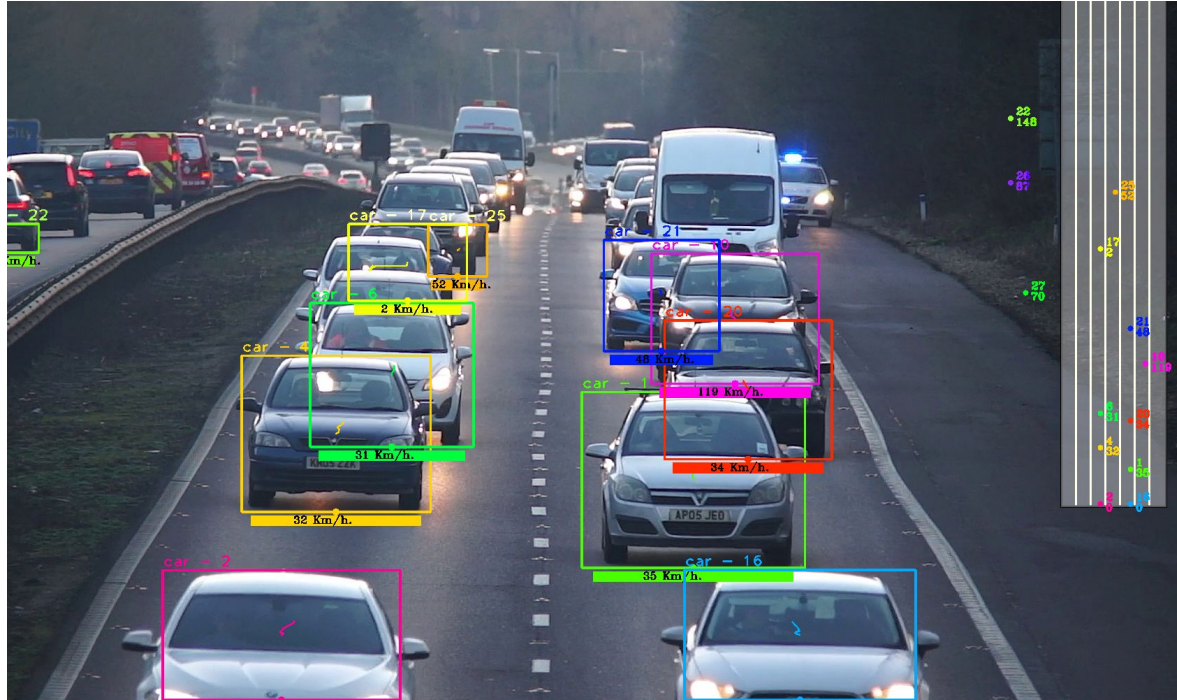


2. Converting Pixel Distance to km/h

- To convert the pixel movement into real-world speed, we use a scaling factor based on a known reference distance in the transformed frame.
- The formula for speed calculation is:

$$\text{Speed (Km/h)} = \left(\frac{\text{Pixel Distance}}{\text{Pixels per Meter}} \right) \times \frac{3600}{\text{FPS} \times 1000}$$

Snap shots





Conclusion:

The traffic monitoring and ticket generation system efficiently detects, tracks, and calculates the speed of vehicles using YOLOv8, Deep SORT, and Kalman Filter. It automates speeding ticket generation, improving traffic enforcement accuracy and contributing to smart city solutions.

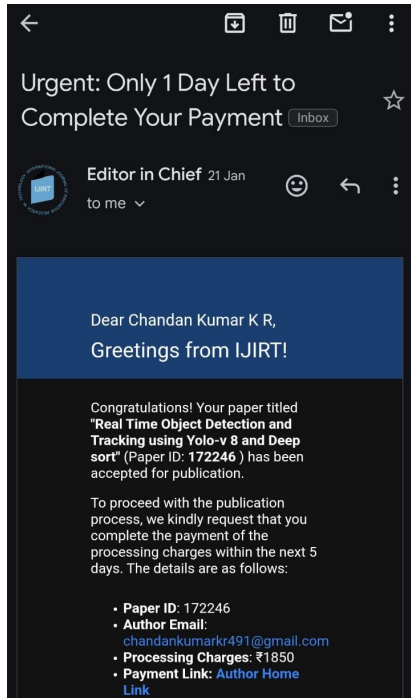
Future Work:

1. Integrate multiple camera feeds for better coverage.
2. Enhance vehicle classification with advanced AI models.
3. Implement real-time alerts for speeding violations.
4. Improve speed estimation with additional data sources.
5. Link the system to traffic control systems for smarter management.



Publication :

our paper has been selected for publication also:



Paper ID : 105

Title : Real Time Object Detection and Tracking using YoloV8 and Deep Sort

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Congratulations! On behalf of the Program Committee of ICTIS 2025 Bangkok, Thailand, I am happy to inform you that your above-mentioned paper has been ACCEPTED for oral presentation in ICTIS 2025, Bangkok, Thailand and publication in Springer LNNS series subject to fulfilment of Guidelines by Springer. An accepted paper will be published in the Springer proceedings (LNNS) only if the final version is accompanied by the payment information (i.e transaction reference number) subject to quality check as per Springer Guidelines.

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